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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth Semester

IOT SYSTEM DESIGN-MG4DSCECT202, MG4DSEECC200

Scheme and Key

Part A: Multiple Choice Questions (MCQs)

1. Which sensor is commonly used to measure temperature? **Answer: a) Thermistor**
2. Which of the following is an example of an IoT gadget? **Answer: a) Smart watch**
3. Smart energy meters used in factories are an example of: **Answer: b) Industrial IoT**
4. Select the IoT device used for monitoring heart rate in healthcare applications. **Answer: b) Pulse sensor**
5. XMPP is primarily used for: **Answer: d) Real-time messaging and presence**
6. CoAP is mainly designed for: **Answer: d) Constrained IoT devices**
7. ESP8266 GPIO pins operate at: **Answer: b) 3.3 V logic level**
8. A gyroscope sensor is used to measure: **Answer: c) Angular velocity**
9. The GPIO pins of Raspberry Pi Zero operate at: **Answer: b) 3.3V logic**
10. Firebase Hosting is mainly used to: **Answer: b) Host web applications**

Part B: Short Answer Questions

11. Define IoT. Explain main components of IoT system.

IoT is a network of connected devices that communicate and exchange data. Main components include sensors/actuators, connectivity (protocols), data processing units, and user interfaces.

12. Explain the working of IoT system.

Sensors collect data, which is transmitted via communication protocols to processing units (like cloud servers), then analyzed and presented to users or trigger actions.

13. Explain privacy issues in IoT systems with examples.

Privacy concerns include data breaches and unauthorized access; e.g., health data leaks

14. Explain Medical IoT (IoMT) and its importance in healthcare.

IoMT improves healthcare through remote monitoring, real-time diagnostics.

15. Explain the working principle of magnetic actuators with examples.

Magnetic actuators operate on magnetic fields to produce motion; e.g., relays or electromagnetic locks.

16. Explain the role of IR and PIR sensors in home automation with examples.

IR sensors detect infrared radiation for proximity; PIR sensors detect motion, used for security lighting and automated systems.

Part C: Essay Questions

16. Explain the basic components of an embedded system with a diagram.

Diagram-5 marks

Explanations-5 marks

Microcontroller/microprocessor, memory, input/output interfaces, and power supply.

17. What changes are needed to convert manual industrial monitoring into IoT-based monitoring?

IoT based industrial monitoring explanations-10 marks

Incorporate sensors, network connectivity, cloud data storage, and real-time dashboards.

18. Features of the ESP32 suitable for IoT applications, including power efficiency, connectivity, and processing.

ESP 32 features over other processors-10 marks

Dual-core processor, Wi-Fi, Bluetooth, low power consumption, rich I/O options.

19. Working of an ultrasonic sensor, distance measurement technique, and applications in IoT.

Definition of Ultrasonic sensors and working-10 marks

Ultrasonic sensors send sound waves, measure time taken for echo to calculate distance, used in obstacle detection, parking systems.